SMART INDIA HACKATHON 2026

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- **Problem Statement ID –**SIH26134
- **Problem Statement Title - Challenges in aligning skill development programs with industry requirements and emerging job market demands**
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-**
- **Team Name –** RealTech

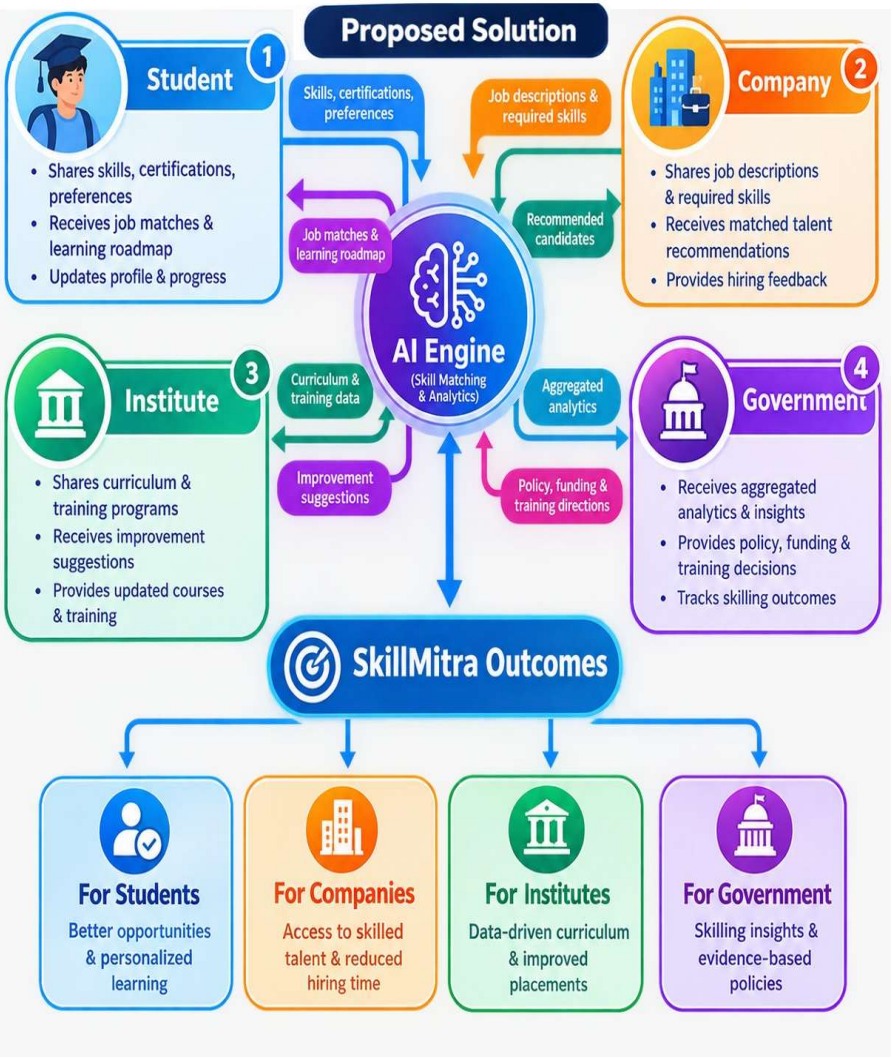


IDEA TITLE –SkillMitra

(AI powered Labour-Market Intelligence & Skill Alignment Platform)

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Proposed Solution



Core Idea

An **AI-powered Labour-Market Intelligence & Skill Alignment Platform** that continuously converts real-time employer demand into evidence-based decisions on **courses, curriculum, trainers, equipment, training capacity** and **candidate career pathways**.

Uniqueness of the Solution

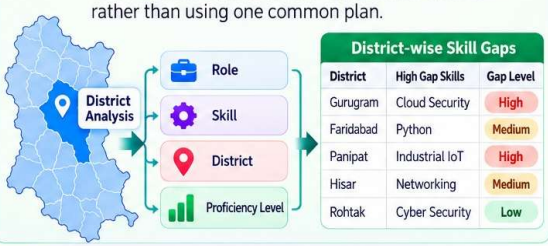
01 Market-to-Curriculum Intelligence

AI doesn't stop at identifying skill gaps—it **maps industry-demanded skills directly to ITI courses, curriculum, trainers and equipment**, showing what training needs to be aligned.



02 District-Level Skill Gap Intelligence

The system analyzes gaps at **role + skill + district + proficiency level**, enabling training plans to be tailored to the actual requirements of a particular district rather than using one common plan.



03 Self-Adaptive Competency & Learning Roadmap

A candidate's competency score continuously changes based on **assessment results, learning progress and practical performance**, allowing the recommended learning path to adapt to the candidate.



04 Closed-Loop Employer Feedback System

The system doesn't end at training or placement. **Placement outcomes and employer feedback become new data**, which continuously improves skill-gap analysis, recommendations and future training plans.



TECHNICAL APPROACH

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TECHNOLOGY STACK
Modern, Scalable and Secure Architecture

01 Frontend
Web application for students, trainers, employers and administrators.

- React / Next.js
- Responsive & modern UI
- Multilingual interface
- Interactive dashboards
- GIS / map visualization

React | NEXT.JS

02 Backend / API
Handles business logic, APIs and AI/ML integration.

- Python + FastAPI
- REST APIs
- Scalable architecture
- Authentication & authorization
- Integration with AI/ML models

Python | FastAPI | API

03 Data & Database
Stores and manages all structured and geospatial data.

- PostgreSQL + PostGIS
- Student, course, job, employer data
- District & geospatial analysis
- High performance & scalable
- Data backup & security

PostgreSQL | PostGIS

04 AI / ML & NLP
Extracts skills, analyzes demand and provides recommendations.

- Python, scikit-learn
- XGBoost / LightGBM
- LLMs (OpenAI / Llama etc.)
- NLP for skill extraction
- Demand forecasting
- Skill gap analysis & recommendations

Python | learn | XGBoost | LightGBM | LLMs

05 Knowledge Graph & Search
Connects skills, roles, courses and industry data for intelligent analysis.

- Neo4j (Knowledge Graph)
- Skill–Role–Task–Course relationships
- Competency mapping
- Elasticsearch / OpenSearch
- Semantic search & skill-based matching
- Industry & curriculum linkage

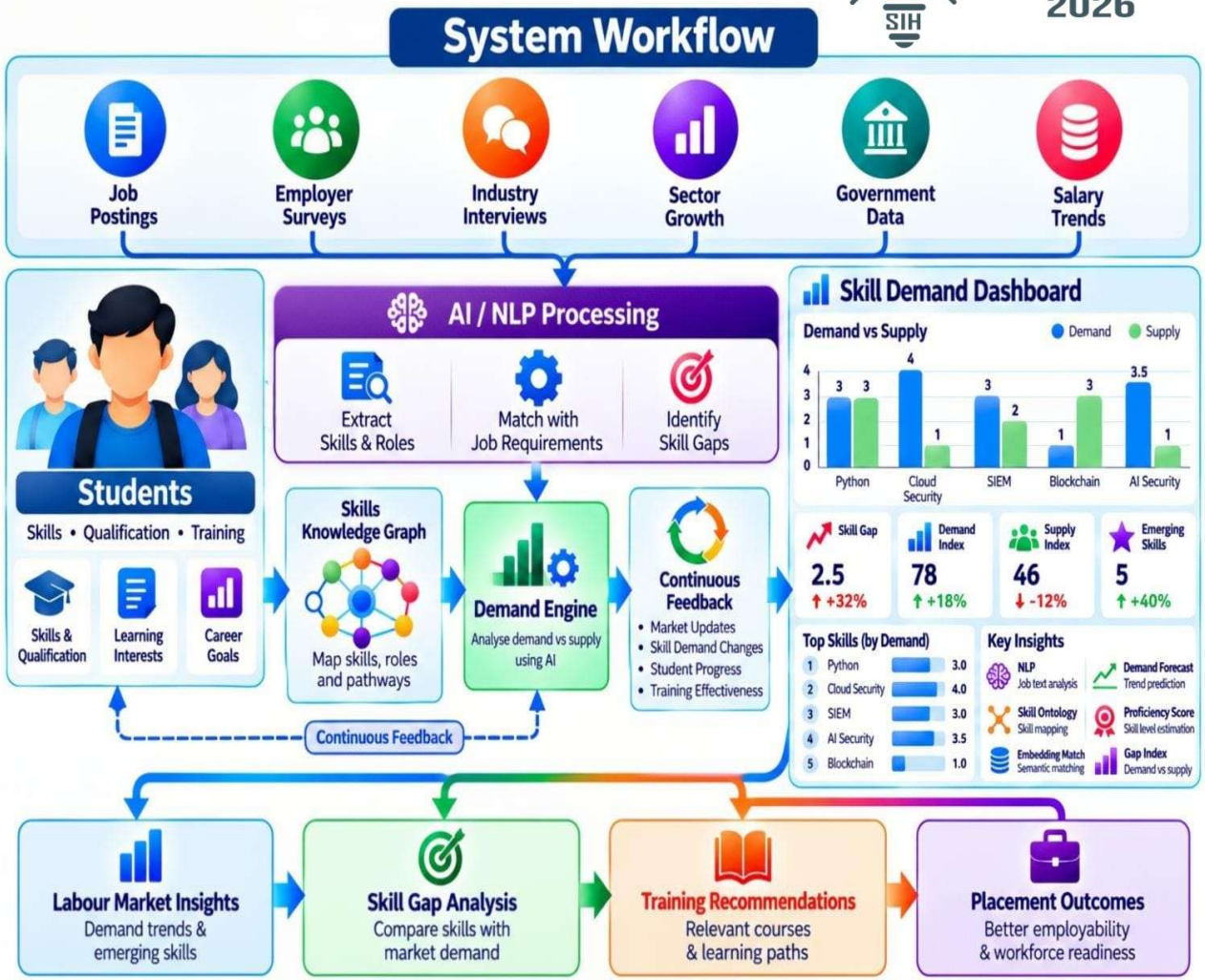
neo4j | elasticsearch | OpenSearch

06 Cloud, Pipeline & Security
Data ingestion, deployment and secure infrastructure.

- Apache Airflow (Batch ETL)
- Apache Kafka (Real-time data)
- Docker (Containerization)
- AWS / Azure / GCP (Cloud Deployment)
- OAuth2 / OIDC (Authentication)
- RBAC (Role Based Access Control)
- Data encryption & audit logs

Airflow | kafka | docker | aws | Azure | Google Cloud

System Workflow



FEASIBILITY AND VIABILITY

FEASIBILITY

Technical Feasibility
- Use of proven AI/NLP models and open-source tools.
- Modular and scalable architecture.
- Integration with existing government systems (e.g., LMS).
- Cloud-ready and secure infrastructure.

Financial Feasibility
- Optimised cost structure with high ROI potential.
- Efficient resource utilisation and budget planning.
- Use of open-source technologies to reduce cost.
- Sustainable model for long-term financial growth.

Scalability & Viability
- Centralised AI platform with modular design for future expansion.
- Applicable across multiple districts and sectors.
- Supports large-scale deployment with minimal additional cost.
- Sustainable and replicable model for long-term impact.

IMPLEMENTATION SUCCESS ROADMAP

Build & Develop
- Define scope & architecture
- Develop core AI/NLP modules
- Pilot with select districts/users
- Prepare training & support material

Integrate
- Connect with existing systems (HR/LMS, job portals, etc.)
- Test & validate
- Ensure smooth adoption

Deploy & Maintain
- Go live with platform
- Monitor performance
- Continuous updates & support
- Scale to more districts/sectors



Revenue Model

One-time
Platform development and initial deployment

Recurring
Maintenance, cloud services and support

Project
Custom implementation for states/departments

CHALLENGES & SOLUTIONS

	CHALLENGES		SOLUTIONS
01	Real-time Job Data Collection	01	**Unified job data aggregation** from multiple sources into a single, clean and structured database.
02	AI Skill Extraction Accuracy	02	**Industry-aligned curriculum** mapping with ITI courses, curriculum, trainers and equipment to recommend updated, relevant training programs.
03	Curriculum Mapping	03	**Automated data cleaning** and validation pipelines to ensure accurate, consistent and reliable data.
04	Keeping Recommendations Updated	04	**Continuous model updates** with new job data and retrain models for up-to-date skill-gap analysis and recommendations.
05	Multi-stakeholder Coordination	05	**Role-based user experience** with custom dashboards and workflows for students, institutes, employers and government departments.
06	Regional Language Support	06	**Multilingual interface** with NLP-powered language processing (English & Marathi) and translated content for a localised user experience.

IMPACTS AND BENEFITS

Potential Impacts

More **students get jobs** after training.

Courses match current **industry needs**.

Better planning for **district skill programs.**

A more **skilled and productive workforce**.

Key Benefits

Stakeholder	Key Benefits with SkillMitra
Students	Better career guidance and job opportunities.
Training Institutes	Updated courses and improved training resources.
Employers	Easier hiring of skilled candidates.
Government	Better decisions using real-time skill demand data.

Stakeholder Network



RESEARCH AND REFERENCES

Policy Frameworks

National Skills Mission REF [01]
District mapping & demand alignment guidelines.

NEP 2020 Reform REF [02]
Multidisciplinary & dynamic syllabus updates.

MahaSwayam Vision REF [03]
State exchange & skill gap analytics framework.

Tech Architecture

NLP Skill Extraction REF [04]
Transformer entity recognition for job & PDF feeds.

GIS Demand Heatmaps REF [05]
Spatial analytics for district shortages & clusters.

Docker Scalability REF [06]
Containerized microservices for secure deployment.

Empirical Data

Live Job Aggregators SRC [01]
Scraping emerging technical & soft-skill demands.

University Repositories SRC [02]
Syllabi documents for automated baseline auditing.

Industry Partner Inputs SRC [03]
Employer metrics for high-fidelity talent matching.